



REC-CIS

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int t;
5     scanf("%d",&t);
6     while(t--)
7     {
8         int n,m,d,min,temp;
9         scanf("%d %d",&n,&m);
10        d=n-m;
11        int arr[n];
12        for (int i=0;i<n;i++)
13            scanf("%d",&arr[i]);
14        for(int j=0;j<n;j++)
15        {
16            min=j;
17            for(int k=j;k<n;k++)
18            {
19                if(arr[k]<arr[min])
20                    min=k;
21            }
22            temp=arr[min];
23            arr[min]=arr[j];
24            arr[j]=temp;
25        }
26        int maxsum=0,minsum=0;
27        for(int a=0;a<d;a++)
28            minsum+=arr[a];
29        for(int b=n-1;b>m-1;b--)
30            maxsum+=arr[b];
31        printf("%d\n",maxsum-minsum);
32    }
33 }
34 }
35 }
```

	Input	Expected	Got	
✓	1 5 1 1 2 3 4 5	4	4	✓

Passed all tests! ✓

Question **2**
Correct
Marked out of 1.00
[Flag question](#)

A new deadly virus has infected large population of a planet. A brilliant scientist has discovered a new strain of virus which can cure this disease. Vaccine produced from this virus has various strength depending on midichlorians count. A person is cured only if midichlorians count in vaccine batch is more than midichlorians count of person. A doctor receives a new set of report which contains midichlorians count of each infected patient, Practo stores all vaccine doctor has and their midichlorians count. You need to determine if doctor can save all patients with the vaccines he has. The number of vaccines and patients are equal.

Input Format

First line contains the number of vaccines - N. Second line contains N integers, which are strength of vaccines. Third line contains N integers, which are midichlorians count of patients.

Output Format

Print a single line containing 'Yes' or 'No'.

Input Constraint

1 < N < 10

Strength of vaccines and midichlorians count of patients fit

100 328 248 689 200

SAMPLE OUTPUT

No

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,min1,min2,temp,flag=1;
5     scanf("%d",&n);
6     int vac[n],pat[n];
7     for(int i=0;i<n;i++)
8         scanf("%d",&vac[i]);
9     for(int i=0;i<n;i++)
10         scanf("%d",&pat[i]);
11
12     for(int j=0;j<n-1;j++)
13     {
14         min1=j,min2=j;
15         for(int k=j;k<n;k++)
16         {
17             if(vac[k]<vac[min1])
18                 min1=k;
19             if(pat[k]<pat[min2])
20                 min2=k;
21
22         }
23         temp=vac[min1];
24         vac[min1]=vac[j];
25         vac[j]=temp;
26
27         temp=pat[min2];
28         pat[min2]=pat[j];
29         pat[j]=temp;
30
31     }
32     for(int i=0;i<n;i++)
33     {
34         if(vac[i]<=pat[i])
35         {
36             flag=0;
37             break;
38         }
39     }
40     if(flag==1)
41         printf("Yes");
42     else
43         printf("No");
44 }
45
```

	Input	Expected	Got	
✓	5 123 146 454 542 456 100 328 248 689 200	No	No	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 1.00

Flag question

You are given an array of n integer numbers $a_1, a_2, \dots, a_n$. Calculate the number of pair of indices (i, j) such that $1 \leq i < j \leq n$ and $a_i \text{ xor } a_j = 0$.

Input format

- First line: n denoting the number of array elements
- Second line: n space separated integers $a_1, a_2, \dots, a_n$.

Output format

Output the required number of pairs.

Constraints

$1 \leq n \leq 10^6$

$1 \leq a_i \leq 10^9$

Output format

Output the required number of pairs.

Constraints

$1 \leq n \leq 10^6$

$1 \leq a_i \leq 10^9$

SAMPLE INPUT

5
1 3 1 4 3

SAMPLE OUTPUT

2

Explanation

The 2 pair of indices are (1, 3) and (2,5).

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,count=0;
5     scanf("%d",&n);
6     int arr[n];
7     for(int i=0;i<n;i++)
8         scanf("%d",&arr[i]);
9     for(int i=0;i<n-1;i++)
10    {
11        for(int j=i+1;j<n;j++)
12        {
13            if((arr[i]^arr[j])==0)
14                count++;
15        }
16    }
17    printf("%d",count);
18 }
```

	Input	Expected	Got	
✓	5 1 3 1 4 3	2	2	✓

Passed all tests! ✓

Question **4**

Correct

Marked out of 1.00

Flag question

You are given an array **A** of non-negative integers of size **m**. Your task is to sort the array in non-decreasing order and print out the original indices of the new sorted array.

Example:

A={4,5,3,7,1}

After sorting the new array becomes A={1,3,4,5,7}.

The required output should be "4 2 0 1 3"

INPUT :

The first line of input consists of the size of the array
The next line consists of the array of size m



Flag question

Example:

A={4,5,3,7,1}

After sorting the new array becomes A={1,3,4,5,7}.

The required output should be "4 2 0 1 3"

INPUT :

The first line of input consists of the size of the array
The next line consists of the array of size m

OUTPUT :

Output consists of a single line of integers

CONSTRAINTS:

1<=m<=106
0<=A[i]<=106

NOTE: The indexing of the array starts with 0.

SAMPLE INPUT

5
4 5 3 7 1

SAMPLE OUTPUT

4 2 0 1 3

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     int arr[n];
7     for(int i=0;i<n;i++)
8         scanf("%d",&arr[i]);
9     int max=arr[0];
10    for(int i=1;i<n;i++)
11    {
12        if(arr[i]>max)
13            max=arr[i];
14    }
15    max++;
16    int min=0;
17    for(int a=0;a<n;a++)
18    {
19        for(int b=0;b<n;b++)
20        {
21            if(arr[b]<arr[min])
22                min=b;
23        }
24        printf("%d ",min);
25        arr[min]=max;
26    }
27 }
28 }
```

	Input	Expected	Got	
✓	5 4 5 3 7 1	4 2 0 1 3	4 2 0 1 3	✓

Passed all tests! ✓

Finish review



GE23131-Programming Using C-2024

Quiz navigation

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3

Show one page at a time

Finish review

Status	Finished
Started	Monday, 13 January 2025, 12:30 PM
Completed	Monday, 13 January 2025, 1:16 PM
Duration	46 mins 43 secs

Question 1

Correct

Marked out of 1.00

Flag question

You are given a two-dimensional 3*3 array starting from A [0][0]. You should add the alternate elements of the array and print its sum. It should print two different numbers the first being sum of A 0 0, A 0 2, A 1 1, A 2 0, A 2 2 and A 0 1, A 1 0, A 1 2, A 2 1.

Input Format

First and only line contains the value of array separated by single space.

A 0 0	A 0 1	A 0 2
4	6	9
A 1 0	A 1 1	A 1 2
2	5	8
A 2 0	A 2 1	A 2 2
1	3	7

Output Format

First line should print sum of A 0 0, A 0 2, A 1 1, A 2 0, A 2 2
Second line should print sum of A 0 1, A 1 0, A 1 2, A 2 1

SAMPLE INPUT

1 2 3 4 5 6 7 8 9

SAMPLE OUTPUT

25
20

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int arr[3][3];
5     for(int i=0;i<3;i++)
6     {
7         for(int j=0;j<3;j++)
8         {
9             scanf("%d",&arr[i][j]);
10        }
11    }
12    int odd=0,even=0;
13    for(int i=0;i<3;i++)
14    {
15        for(int j=0;j<3;j++)
16        {
17            if((i+j)%2!=0)
18                odd+=arr[i][j];
19            else
20                even+=arr[i][j];
21        }
22    }
23    printf("%d\n%d",even,odd);
24 }
```

	Input	Expected
--	-------	----------



REC-CIS

```
20         even+=arr[i][j];
21     }
22 }
23 printf("%d\n%d",even,odd);
24 }
```

	Input	Expected
✓	1 2 3 4 5 6 7 8 9	25 20
✓	21 422 423 443 586 645 657 846 904	2591 2356

Passed all tests! ✓

Question **2**

Correct

Marked out of
5.00

Flag
question

Microsoft has come to hire interns from your college. N students got shortlisted out of which few were males and a few females. All the students have been assigned talent levels. Smaller the talent level, lesser is your chance to be selected. Microsoft wants to create the result list where it wants the candidates sorted according to their talent levels, but there is a catch. This time Microsoft wants to hire female candidates first and then male candidates.

The task is to create a list where first all-female candidates are sorted in a descending order and then male candidates are sorted in a descending order.

Input Format

The first line contains an integer N denoting the number of students. Next, N lines contain two space-separated integers, ai and bi.

The first integer, ai will be either 1(for a male candidate) or 0(for female candidate).

The second integer, bi will be the candidate's talent level.

Constraints

$1 \leq N \leq 10^5$
 $0 \leq ai \leq 1$
 $1 \leq bi \leq 10^9$

Output Format

Output space-separated integers, which first contains the talent levels of all female candidates sorted in descending order and then the talent levels of male candidates in descending order.

SAMPLE INPUT

5
0 3
1 6
0 2
0 7
1 15

SAMPLE OUTPUT

7 3 2 15 6



1 6
0 2
0 7
1 15

SAMPLE OUTPUT

7 3 2 15 6

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 struct data
3 {
4     int gen;int tal;
5 }
6
7 int main()
8 {
9     int n;
10    scanf("%d",&n);
11    struct data a[n];
12    for(int i=0;i<n;i++)
13        scanf("%d %d",&a[i].gen,&a[i].tal);
14    for(int i=0;i<n-1;i++)
15    {
16        for(int j=0;j<n-i-1;j++)
17        {
18            if(a[j].tal<a[j+1].tal)
19            {
20                struct data temp=a[j];
21                a[j]=a[j+1];
22                a[j+1]=temp;
23            }
24        }
25    }
26    for(int i=0;i<n;i++)
27    {
28        if(a[i].gen==0)
29            printf("%d ",a[i].tal);
30    }
31    for(int i=0;i<n;++i)
32    {
33        if(a[i].gen==1)
34            printf("%d ",a[i].tal);
35    }
36 }
37
38 }
```

	Input	Expected	
✓	5 0 3 1 6 0 2 0 7 1 15	7 3 2 15 6	7
✓	6 0 1 0 26 0 39 0 37 0 7 0 13	39 37 26 13 7 1	3
✓	12 1 12 1 14 1 18 1 1 1 2 1 3 1 5 1 8 1 9 1 10 0 29 0 31	31 29 18 14 12 10 9 8 5 3 2 1	3
✓	12 0 12 1 12 0 12 1 12 0 12 0 12 1 12 0 12 1 12 1 12 0 12	12 12 12 12 12 12 12 12 12 12 12 12	1



SAMPLE OUTPUT

7 3 2 15 6

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 struct data
3 {
4     int gen;int tal;
5 };
6
7 int main()
8 {
9     int n;
10    scanf("%d",&n);
11    struct data a[n];
12    for(int i=0;i<n;i++)
13        scanf("%d %d",&a[i].gen,&a[i].tal);
14    for(int i=0;i<n-1;i++)
15    {
16        for(int j=0;j<n-i-1;j++)
17        {
18            if(a[j].tal<a[j+1].tal)
19            {
20                struct data temp=a[j];
21                a[j]=a[j+1];
22                a[j+1]=temp;
23            }
24        }
25    }
26    for(int i=0;i<n;i++)
27    {
28        if(a[i].gen==0)
29            printf("%d ",a[i].tal);
30    }
31    for(int i=0;i<n;++i)
32    {
33        if(a[i].gen==1)
34            printf("%d ",a[i].tal);
35    }
36 }
37
38 }
```

	Input	Expected	
✓	5 0 3 1 6 0 2 0 7 1 15	7 3 2 15 6	7
✓	6 0 1 0 26 0 39 0 37 0 7 0 13	39 37 26 13 7 1	3
✓	12 1 12 1 14 1 18 1 1 1 2 1 3 1 5 1 8 1 9 1 10 0 29 0 31	31 29 18 14 12 10 9 8 5 3 2 1	3
✓	12 0 12 1 12 0 12 1 12 0 12 0 12 1 12 0 12 1 12 1 12 0 12 1 12	12 12 12 12 12 12 12 12 12 12 12 12 1	1

Passed all tests! ✓

Question 3

Correct

Marked out of 1.00

Flag question

Shyam Lal, a wealthy landlord from the state of Rajasthan, being an old fellow and tired of doing hard work, decided to sell all his farmland and to live rest of his life with that money. No other farmer is rich enough to buy all his land so he decided to partition the land into rectangular plots of different sizes with different cost per unit area. So, he sold these plots to the farmers but made a mistake. Being illiterate, he made partitions that could be overlapping. When the farmers came to know about it, they ran to him for compensation of extra money they paid to him. So, he decided to return all the money to the farmers of that land which was overlapping with other farmer's land to settle down the conflict. All the portion of conflicted land will be taken back by the landlord.

To decide the total compensation, he has to calculate the total amount of money to return back to farmers with the same cost they had purchased from him. Suppose, Shyam Lal has a total land area of **1000 x 1000** equal square blocks where each block is equivalent to a unit square area which can be represented on the co-ordinate axis. Now find the total amount of money, he has to return to the farmers. Help Shyam Lal to accomplish this task.

Input Format:

The first line of the input contains an integer **N**, denoting the total number of land pieces he had distributed. Next **N** line contains the **5** space separated integers (**X1, Y1**), (**X2, Y2**) to represent a rectangular piece of land, and cost per unit area **C**.

(**X1, Y1**) and (**X2, Y2**) are the locations of first and last square block on the diagonal of the rectangular region.

Output Format:

Print the total amount he has to return to farmers to solve the conflict.

Constraints:

- $1 \leq N \leq 100$
- $1 \leq X1 \leq X2 \leq 1000$
- $1 \leq Y1 \leq Y2 \leq 1000$
- $1 \leq C \leq 1000$

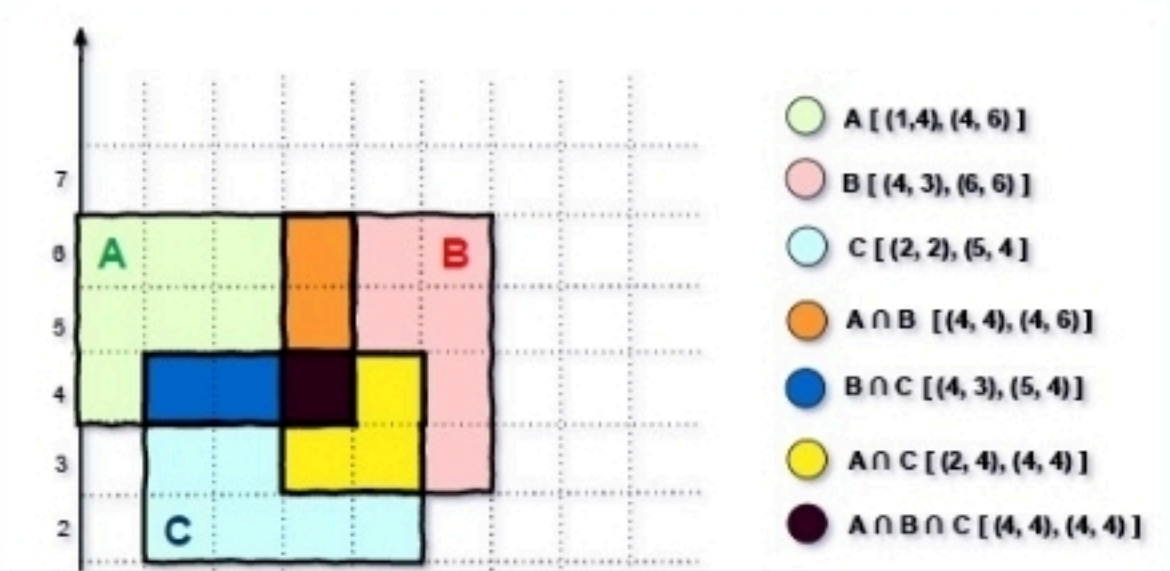
SAMPLE INPUT

3
1 4 4 6 1
4 3 6 6 2
2 2 5 4 3

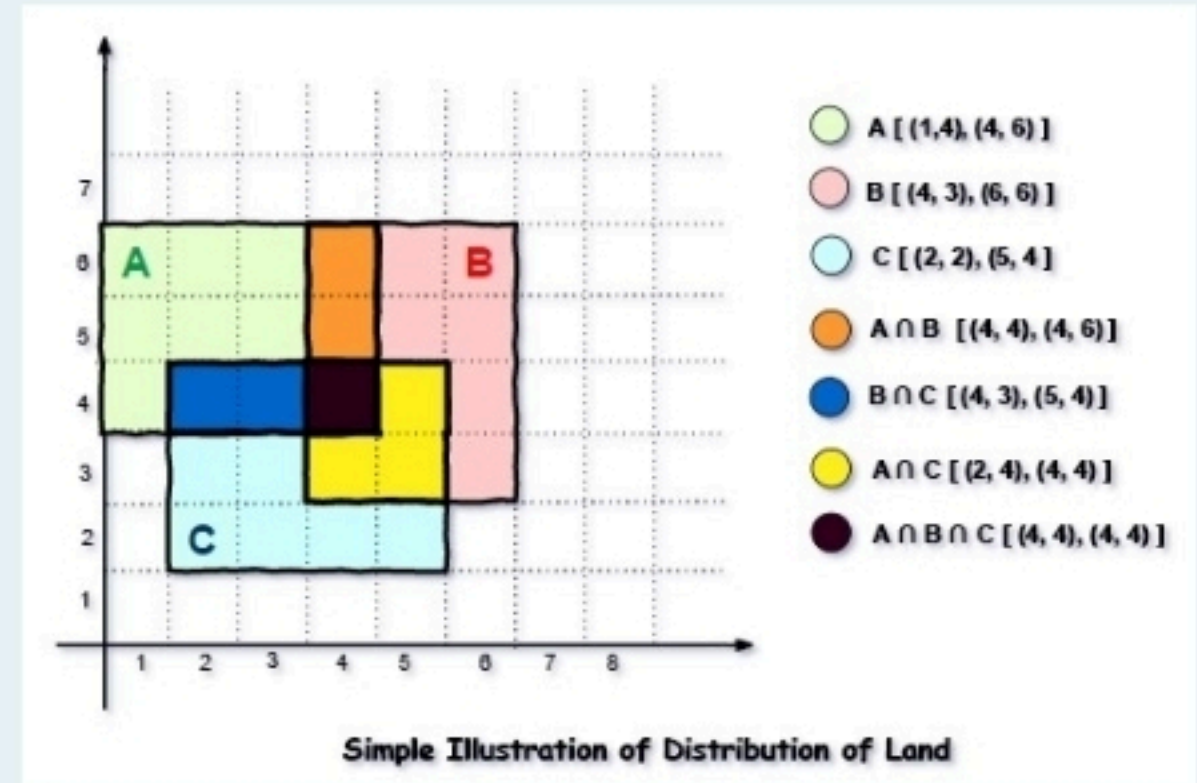
SAMPLE OUTPUT

35

Explanation



Explanation



For given sample input (see given graph for reference), compensation money for different farmers is as follows:

Farmer with land area A: $C_1 = 5 * 1 = 5$

Farmer with land area B: $C_2 = 6 * 2 = 12$

Farmer with land area C: $C_3 = 6 * 3 = 18$

Total Compensation Money = $C_1 + C_2 + C_3 = 5 + 12 + 18 = 35$

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,n,x1,x2,y1,y2,t=0;
5     long long total=0;
6     int arr[1001][1001]={0};
7     scanf("%d",&n);
8     while(n--)
9     {
10         scanf("%d %d %d %d %d",&x1,&y1,&x2,&y2,&t);
11         for(i=x1;i<=x2;i++)
12         {
13             for(j=y1;j<=y2;j++)
14             {
15                 if(arr[i][j]==0)
16                     arr[i][j]+=t;
17                 else if(arr[i][j]>0)
18                     arr[i][j]=(-1)*(arr[i][j]);
19                 else if(arr[i][j]<0)
20                     arr[i][j]-=t;
21             }
22         }
23     }
24     for(i=1;i<1001;i++)
25     {
26         for(j=1;j<1001;j++)
27         {
28             if(arr[i][j]<0)
29                 total+=arr[i][j];
30         }
31     }
32     printf("%lld\n",(-1)*total);
33     return 0;
34 }
35 }
```

	Input	Expected	Got	
✓	3 1 4 4 6 1 4 3 6 6 2 2 2 5 4 3	35	35	✓
✓	1 48 12 49 27 8	0	0	✓
✓	3 88 34 99 76 44 82 65 94 100 81 58 16 65 66 7	10500	10500	✓

Passed all tests! ✓



GE23131-Programming Using C-2024

Quiz navigation

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Show one page at a time

Finish review

Status	Finished
Started	Monday, 13 January 2025, 4:24 PM
Completed	Monday, 13 January 2025, 4:52 PM
Duration	28 mins 18 secs

Question 1

Correct

Marked out of 1.00

Flag question

Given a string, *s*, consisting of alphabets and digits, find the frequency of each digit in the given string.

Input Format

The first line contains a string, *num* which is the given number.

Constraints

$1 \leq len(num) \leq 1000$

All the elements of num are made of English alphabets and digits.

Output Format

Print ten space-separated integers in a single line denoting the frequency of each digit from 0 to 9.

Sample Input 0

a11472o5t6

Sample Output 0

0 2 1 0 1 1 1 1 0 0

Explanation 0

In the given string:

- 1 occurs two times.
 - 2, 4, 5, 6 and 7 occur one time each.
- The remaining digits 0, 3, 8 and 9 don't occur at all.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     char str[1000];
5     scanf("%s",str);
6     int hash[10]={0,0,0,0,0,0,0,0,0,0};
7     int temp;
8     for(int i=0;str[i]!='\0';i++)
9     {
10         temp=str[i]-'0';
11         if(temp<=9&&temp>=0)
12         {
13             hash[temp]++;
14         }
15     }
16     for(int i=0;i<=9;i++)
17     {
18         printf("%d ",hash[i]);
19     }
20     return 0;
21 }
```



REC-CIS

Print ten space-separated integers in a single line denoting the frequency of each digit from **0** to **9**.

Sample Input 0

a11472o5t6

Sample Output 0

0 2 1 0 1 1 1 1 0 0

Explanation 0

In the given string:

- **1** occurs two times.
 - **2, 4, 5, 6** and **7** occur one time each.
- The remaining digits **0, 3, 8** and **9** don't occur at all.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     char str[1000];
5     scanf("%s",str);
6     int hash[10]={0,0,0,0,0,0,0,0,0,0};
7     int temp;
8     for(int i=0;str[i]!='\0';i++)
9     {
10         temp=str[i]-'0';
11         if(temp<=9&&temp>=0)
12         {
13             hash[temp]++;
14         }
15     }
16     for(int i=0;i<=9;i++)
17     {
18         printf("%d ",hash[i]);
19     }
20     return 0;
21 }
```

	Input	Expected	G
✓	a11472o5t6	0 2 1 0 1 1 1 1 0 0	0
✓	lw4n88j12n1	0 2 1 0 1 0 0 0 2 0	0
✓	1v888861256338ar0ekk	1 1 1 2 0 1 2 0 5 0	1

Passed all tests! ✓

Question **2**

Correct

Marked out of 1.00

 Flag question

Today, Monk went for a walk in a garden. There are many trees in the garden and each tree has an English alphabet on it. While Monk was walking, he noticed that all trees with vowels on it are not in good state. He decided to take care of them. So, he asked you to tell him the count of such trees in the garden.

Note: The following letters are vowels: 'A', 'E', 'I', 'O', 'U', 'a', 'e', 'i', 'o' and 'u'.

Input:

The first line consists of an integer *T* denoting the number of test cases.

Each test case consists of only one string, each character of string denoting the alphabet (may be lowercase or uppercase) on a tree in the garden

Question 2

Correct

Marked out of 1.00

Flag question

Today, Monk went for a walk in a garden. There are many trees in the garden and each tree has an English alphabet on it. While Monk was walking, he noticed that all trees with vowels on it are not in good state. He decided to take care of them. So, he asked you to tell him the count of such trees in the garden.

Note: The following letters are vowels: 'A', 'E', 'I', 'O', 'U', 'a', 'e', 'i', 'o' and 'u'.

Input:

The first line consists of an integer T denoting the number of test cases.

Each test case consists of only one string, each character of string denoting the alphabet (may be lowercase or uppercase) on a tree in the garden.

Output:

For each test case, print the count in a new line.

Constraints:

$1 \leq T \leq 10$
 $1 \leq \text{length of string} \leq 10^5$

SAMPLE INPUT

2
nBBZLaosnm
JHkIsnZtTL

SAMPLE OUTPUT

2
1

Explanation

In test case 1, a and o are the only vowels. So, count=2

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int t;
5     scanf("%d",&t);
6     while(t--)
7     {
8         char str[100000];
9         int count=0;
10        scanf("%s",str);
11        for(int i=0;str[i]!='\0';i++)
12        {
13            char c= str[i];
14            if((c=='a')||(c=='e')||(c=='i'
15            count++;
16        }
17        printf("%d\n",count);
18    }
19    return 0;
20 }
```

	Input	Expected	Got	
✓	2 nBBZLaosnm JHkIsnZtTL	2 1	2 1	✓
✓	2 nBBZLaosnm JHkIsnZtTL	2 1	2 1	✓

Passed all tests! ✓



REC-CIS

Question **3**

Correct

Marked out of 1.00

Flag question

Given a sentence, **s**, print each word of the sentence in a new line.

Input Format

The first and only line contains a sentence, **s**.

Constraints

$1 \leq len(s) \leq 1000$

Output Format

Print each word of the sentence in a new line.

Sample Input 0

This is C

Sample Output 0

This
is
C

Explanation 0

In the given string, there are three words ["This", "is", "C"]. We have to print each of these words in a new line.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     char s[1000];
5     scanf("%[^\n]s",s);
6     for(int i=0;s[i]!='\0';i++)
7     {
8         if (s[i]!=' ')
9             printf("%c",s[i]);
10        else
11            printf("\n");
12    }
13    return 0;
14 }
```

	Input	Expected	Got	
✓	This is C	This is C	This is C	✓
✓	Learning C is fun	Learning C is fun	Learning C is fun	✓

Passed all tests! ✓

Question **4**

Correct

Marked out of

Input Format

Question **4**

Correct

Marked out of 1.00

[Flag question](#)

Input Format

You are given two strings, **a** and **b**, separated by a new line. Each string will consist of lower case Latin characters ('a'- 'z').

Output Format

In the first line print two space-separated integers, representing the length of **a** and **b** respectively.

In the second line print the string produced by concatenating **a** and **b** (**a + b**).

In the third line print two strings separated by a space, **a'** and **b'**. **a'** and **b'** are the same as **a** and **b**, respectively, except that their first characters are swapped.

Sample Input

abcd
ef

Sample Output

4 2
abcdef
ebcd af

Explanation

a = "abcd"
b = "ef"
|a| = 4
|b| = 2
a + b = "abcdef"
a' = "ebcd"
b' = "af"

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     char str1[10],str2[10],t;
5     int i=0,j=0;
6     int count1=0,count2=0;
7     scanf("%s",str1);
8     scanf("%s",str2);
9     while(str1[i]!='\0')
10    {
11        count1++;
12        i++;
13    }
14    while(str2[j]!='\0')
15    {
16        count2++;
17        j++;
18    }
19    printf("%d %d\n",count1,count2);
20    printf("%s%s\n",str1,str2);
21    t=str1[0];
22    str1[0]=str2[0];
23    str2[0]=t;
24    printf("%s %s",str1,str2);
25    return 0;
26 }
27
28
```

	Input	Expected	Got	
✓	abcd ef	4 2 abcdef ebcd af	4 2 abcdef ebcd af	✓

Passed all tests! ✓



REC-CIS

Quiz navigation

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Finish review

Status	Finished
Started	Monday, 13 January 2025, 4:53 PM
Completed	Monday, 13 January 2025, 5:58 PM
Duration	1 hour 4 mins

Question 1

Correct

Marked out of 1.00

Flag question

Two strings **A** and **B** comprising of lower case English letters are compatible if they are equal or can be made equal by following this step any number of times:

- Select a prefix from the string **A** (possibly empty), and increase the alphabetical value of all the characters in the prefix by the same valid amount. For example, if the string is **xyz** and we select the prefix **xy** then we can convert it to **yx** by increasing the alphabetical value by 1. But if we select the prefix **xyz** then we cannot increase the alphabetical value.

Your task is to determine if given strings **A** and **B** are compatible.

Input format

First line: String **A**
Next line: String **B**

Output format

For each test case, print **YES** if string **A** can be converted to string **B**, otherwise print **NO**.

Constraints

$1 \leq len(A) \leq 1000000$
 $1 \leq len(B) \leq 1000000$

SAMPLE INPUT

abaca
cdbda

SAMPLE OUTPUT

YES

Explanation

The string **abaca** can be converted to **bcbda** in one move and to **cdbda** in the next move.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     char str1[1000000],str2[1000000];
6     int flag=1;
7     scanf("%s",str1);
8     scanf("%s",str2);
9     int a=strlen(str1);
10    int b=strlen(str2);
11    if(a==b)
12    {
13        for(int i=a-1;i>=0;i--)
14        {
15            while(str1[i]!=str2[i])
16            {
17                for(int j=0;j<=i;j++)
18                {
```




Explanation

The string **abaca** can be converted to **bcdba** in one move and to **cdbda** in the next move.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     char str1[1000000],str2[1000000];
6     int flag=1;
7     scanf("%s",str1);
8     scanf("%s",str2);
9     int a=strlen(str1);
10    int b=strlen(str2);
11    if(a==b)
12    {
13        for(int i=a-1;i>=0;i--)
14        {
15            while(str1[i]!=str2[i])
16            {
17                for(int j=0;j<=i;j++)
18                {
19                    if(str1[j]<'z')
20                        str1[j]++;
21                    else
22                    {
23                        flag=0;
24                        break;
25                    }
26                }
27                if(flag==0)
28                    break;
29            }
30        }
31    }
32    else
33        flag=0;
34
35    if(flag==0)
36        printf("NO");
37    else
38        printf("YES");
39    return 0;
40 }
```

	Input	Expected	Got	
✓	abaca cdbda	YES	YES	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 1.00

Flag question

Danny has a possible list of passwords of Manny's facebook account. All passwords length is odd. But Danny knows that Manny is a big fan of palindromes. So, his password and reverse of his password both should be in the list.

You have to print the length of Manny's password and it's middle character.

Note: The solution will be unique.

INPUT

The first line of input contains the integer N, the number of possible passwords.

Each of the following N lines contains a single word, its length being an odd number greater than 2 and lesser than 14. All characters are lowercase letters of the English alphabet.

OUTPUT

The first and only line of output must contain the length of the correct password and its central letter.

CONSTRAINTS



Passed all tests! ✓

Question 2

Correct

Marked out of 1.00

Flag question

Danny has a possible list of passwords of Manny's facebook account. All passwords length is odd. But Danny knows that Manny is a big fan of palindromes. So, his password and reverse of his password both should be in the list.

You have to print the length of Manny's password and it's middle character.

Note: The solution will be unique.

INPUT

The first line of input contains the integer N, the number of possible passwords.

Each of the following N lines contains a single word, its length being an odd number greater than 2 and lesser than 14. All characters are lowercase letters of the English alphabet.

OUTPUT

The first and only line of output must contain the length of the correct password and its central letter.

CONSTRAINTS

$1 \leq N \leq 100$

SAMPLE INPUT

4
abc
def
feg
cba

SAMPLE OUTPUT

3 b

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     int n,flag=0;
6     char temp;
7     scanf("%d",&n);
8     char words[n][14];
9     for(int i=0;i<n;i++)
10     scanf("%s",words[i]);
11     char reverse[14];
12     for(int i=0;i<n-1;i++)
13     {
14         strcpy(reverse,words[i]);
15         reverse[0]='\0';
16         reverse[size-1]='\0';
17         for(int k=0;k<size/2;k++)
18         {
19             temp=reverse[k];
20             reverse[k]=reverse[size-k-1];
21             reverse[size-k-1]=temp;
22         }
23     }
24     for(int j=i+1;j<n;j++)
25     {
26         if(strcmp(reverse,words[j])==0)
27         {
28             flag=1;
29             break;
30         }
31     }
32     if(flag==1)
33     {
34         printf("%d %c\n",size,reverse[size/2]);
35     }
36 }
```

200%



Ultra volume mode on

Joey has assigned points to all the restaurants, but can't figure out which restaurant satisfies Chandler's criteria. Can you help him out?

Input:

First line has N, the total number of restaurants.

Next N lines contain Name of Restaurant and Points awarded by Joey, separated by a space. Restaurant name has **no spaces**, all lowercase letters and will not be more than 20 characters.

Output:

Print the name of the restaurant that Joey should choose.

Constraints:

1 <= N <= 10⁵

1 <= Points <= 10⁶

SAMPLE INPUT

3
Pizzeria 108
Dominos 145
Pizzapizza 49

SAMPLE OUTPUT

Dominos

Explanation

Dominos has maximum points.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     int n;
6     scanf("%d",&n);
7     char res[n][21];
8     int rate[n];
9     for(int i=0;i<n;i++)
10    {
11        scanf("%s",res[i]);
12        scanf("%d",&rate[i]);
13    }
14    int max=rate[0];
15    char ans[20];
16    strcpy(ans,res[0]);
17    for(int i=1;i<n;i++)
18    {
19        if(rate[i]>max)
20        {
21            max=rate[i];
22            strcpy(ans,res[i]);
23        }
24        else if(rate[i]==max)
25        {
26            if(strcmp(res[i],ans)<0)
27                strcpy(ans,res[i]);
28        }
29    }
30    printf("%s",ans);
31    return 0;
32 }
```

	Input	Expected	Got	
✓	3 Pizzeria 108 Dominos 145 Pizzapizza 49	Dominos	Dominos	✓

SAMPLE OUTPUT

3 b

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     int n,flag=0;
6     char temp;
7     scanf("%d",&n);
8     char words[n][14];
9     for(int i=0;i<n;i++)
10     scanf("%s",words[i]);
11     char reverse[14];
12     for(int i=0;i<n-1;i++)
13     {
14         strcpy(reverse,words[i]);
15         int size=strlen(reverse);
16
17         for(int k=0;k<size/2;k++)
18         {
19             temp=reverse[k];
20             reverse[k]=reverse[size-k-1];
21             reverse[size-k-1]=temp;
22         }
23         for(int j=i+1;j<n;j++)
24         {
25             if(strcmp(reverse,words[j])==0)
26             {
27                 flag=1;
28                 break;
29             }
30         }
31         if(flag==1)
32             break;
33     }
34     int len=strlen(reverse);
35     printf("%d %c ",len,reverse[len/2]);
36     return 0;
37 }
```

	Input	Expected	Got	
✓	4 abc def feg cba	3 b	3 b	✓

Passed all tests! ✓

Question **3**

Correct

Marked out of 1.00

Flag question

Joey loves to eat Pizza. But he is worried as the quality of pizza made by most of the restaurants is deteriorating. The last few pizzas ordered by him did not taste good :(. Joey is feeling extremely hungry and wants to eat pizza. But he is confused about the restaurant from where he should order. As always he asks Chandler for help.

Chandler suggests that Joey should give each restaurant some points, and then choose the restaurant having **maximum points**. If more than one restaurant has same points, Joey can choose the one with **lexicographically smallest** name.

Joey has assigned points to all the restaurants, but can't figure out which restaurant satisfies Chandler's criteria. Can you help him out?

Input:

First line has N, the total number of restaurants.

Next N lines contain Name of Restaurant and Points awarded by Joey, separated by a space. Restaurant name has **no spaces**, all lowercase letters and will not be more than 20 characters.



Question **4**

Correct

Marked out of 1.00

Flag question

These days Bechan Chacha is depressed because his crush gave him list of mobile number some of them are valid and some of them are invalid. Bechan Chacha has special power that he can pick his crush number only if he has valid set of mobile numbers. Help him to determine the valid numbers.

You are given a string "S" and you have to determine whether it is Valid mobile number or not. Mobile number is valid only if it is of length 10 , consists of numeric values and it shouldn't have prefix zeroes.

Input:

First line of input is T representing total number of test cases.
Next T line each representing "S" as described in in problem statement.

Output:

Print "YES" if it is valid mobile number else print "NO".
Note: Quotes are for clarity.

Constraints:

1<= T <= 10³
sum of string length <= 10⁵

SAMPLE INPUT

3
1234567890
0123456789
0123456.87

SAMPLE OUTPUT

YES
NO
NO

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     int t;
6     scanf("%d",&t);
7     while(t--)
8     {
9         int flag=1;
10        char s[100000];
11        scanf("%s",s);
12        int k=strlen(s);
13
14        if(k==10)
15        {
16            for(int i=0;i<10;i++)
17            {
18                if(s[i]!='0')
19                {
20                    flag=0;
21                    break;
22                }
23                if(s[i]<'0' || s[i]>'9')
24                {
25                    flag=0;
26                    break;
27                }
28            }
29        }
30        else
31        flag=0;
32        if(flag==1)
33        printf("YES\n");
34        else
35        printf("NO\n");
36    }
37 }
38 }
```

statement.

Output:

Print "YES" if it is valid mobile number else print "NO".
Note: Quotes are for clarity.

Constraints:

1<= T <= 10³
sum of string length <= 10⁵

SAMPLE INPUT

3
1234567890
0123456789
0123456.87

SAMPLE OUTPUT

YES
NO
NO

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<string.h>
3 int main()
4 {
5     int t;
6     scanf("%d",&t);
7     while(t--)
8     {
9         int flag=1;
10        char s[100000];
11        scanf("%s",s);
12        int k=strlen(s);
13
14        if(k==10)
15        {
16            for(int i=0;i<10;i++)
17            {
18                if(s[i]!='0')
19                {
20                    flag=0;
21                    break;
22                }
23                if(s[i]<'0' || s[i]>'9')
24                {
25                    flag=0;
26                    break;
27                }
28            }
29        }
30        else
31        {
32            flag=0;
33            if(flag==1)
34                printf("YES\n");
35            else
36                printf("NO\n");
37        }
38    }
39    return 0;
40 }
```

	Input	Expected	Got	
✓	3	YES	YES	✓
	1234567890	NO	NO	
	0123456789	NO	NO	
	0123456.87			

Passed all tests! ✓



REC-CIS

Sample Output 1

1

Explanation 1

- Convert the decimal number 77 to binary number: $77_{10} = (1001101)_2$.
- The value of the 4th index from the right in the binary representation is 1.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2   * Complete the 'fourthBit' function below
3   *
4   * The function is expected to return an
5   * The function accepts INTEGER number as
6   */
7
8  int fourthBit(int number)
9  {
10     int binary[32];
11     int i=0;
12     while(number>0)
13     {
14         binary[i]=number%2;
15         number/=2;
16         i++;
17     }
18     if(i>=4)
19     {
20         return binary[3];
21     }
22     else
23         return 0;
24
25 }
```

	Test	Expected	Got
✓	printf("%d", fourthBit(32))	0	0
✓	printf("%d", fourthBit(77))	1	1

Passed all tests! ✓

Question 2

Correct

Marked out of 1.00

Flag question

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the p^{th} element of the list, sorted ascending. If there is no p^{th} element, return 0.

Example

n = 20
p = 3

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if p = 3, then 4 is returned. If p > 6, 0 would be returned.

Function Description

Complete the function pthFactor in the editor below.

pthFactor has the following parameter(s):

- int n: the integer whose factors are to be found
- int p: the index of the factor to be returned

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Finish review

Status	Finished
Started	Monday, 13 January 2025, 6:15 PM
Completed	Monday, 13 January 2025, 6:22 PM
Duration	6 mins 44 secs

Question 1

Correct

Marked out of 1.00

Flag question

A binary number is a combination of 1s and 0s. Its n^{th} least significant digit is the n^{th} digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the the 4^{th} least significant digit.

Example

number = 23

- Convert the decimal number 23 to binary number: $23^{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$.
- The value of the 4^{th} index from the right in the binary representation is 0.

Function Description

Complete the function fourthBit in the editor below.

fourthBit has the following parameter(s):

int number: a decimal integer

Returns:

int: an integer 0 or 1 matching the 4th least significant digit in the binary representation of number.

Constraints

$0 \leq \text{number} < 2^{31}$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The only line contains an integer, number.

Sample Case 0

Sample Input 0

STDIN Function

32 → number = 32

Sample Output 0

0

Explanation 0

- Convert the decimal number 32 to binary number: $32_{10} = (100000)_2$.

Sample Input 1

STDIN Function

10 → n = 10
5 → p = 5

Sample Output 1

0

Explanation 1

Factoring n = 10 results in {1, 2, 5, 10}. There are only 4 factors and p = 5, therefore 0 is returned as the answer.

Sample Case 2

Sample Input 2

STDIN Function

1 → n = 1
1 → p = 1

Sample Output 2

1

Explanation 2

Factoring n = 1 results in {1}. The p = 1st factor of 1 is returned as the answer.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2  * Complete the 'pthFactor' function below
3  *
4  * The function is expected to return a LONG_INTEGER
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER n
7  * 2. LONG_INTEGER p
8  */
9
10 long pthFactor(long n, long p)
11 {
12     int count=0;
13     for(long i=1;i<=n;++i)
14     {
15         if(n%i==0)
16         {
17             count++;
18             if(count==p)
19             {
20                 return i;
21             }
22         }
23     }
24     return 0;
25 }
26
```

	Test	Expected	Got
✓	printf("%ld", pthFactor(10, 3))	5	5
✓	printf("%ld", pthFactor(10, 5))	0	0
✓	printf("%ld", pthFactor(1, 1))	1	1

Passed all tests! ✓

Finish review



REC-CIS



Show one page at a time

Finish review

Status	Finished
Started	Monday, 13 January 2025, 7:41 PM
Completed	Monday, 13 January 2025, 7:49 PM
Duration	8 mins 9 secs

Question **1**

Correct

Marked out of 1.00

Flag question

You are a bank account hacker. Initially you have 1 rupee in your account, and you want exactly **N** rupees in your account. You wrote two hacks, first hack can multiply the amount of money you own by 10, while the second can multiply it by 20. These hacks can be used any number of time. Can you achieve the desired amount **N** using these hacks.

Constraints:

1<=T<=100
1<=N<=10^12

Input

The test case contains a single integer N.

Output

For each test case, print a single line containing the string "1" if you can make exactly N rupees or "0" otherwise.

SAMPLE INPUT

1

SAMPLE OUTPUT

1

SAMPLE INPUT

2

SAMPLE OUTPUT

0

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2   * Complete the 'myFunc' function below.
3   *
4   * The function is expected to return an
5   * The function accepts INTEGER n as para
6   */
7
8  int myFunc(int n)
9  {
10     return n==1 || n%10==0;
11 }
12
13
```


Output

For each test case, print a single line containing the string "1" if you can make exactly N rupees or "0" otherwise.

SAMPLE INPUT

1

SAMPLE OUTPUT

1

SAMPLE INPUT

2

SAMPLE OUTPUT

0

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2   * Complete the 'myFunc' function below.
3   *
4   * The function is expected to return an integer.
5   * The function accepts INTEGER n as parameter.
6   */
7
8  int myFunc(int n)
9  {
10     return n==1 || n%10==0;
11 }
12
13
```

	Test	Expected	Got	
✓	printf("%d", myFunc(1))	1	1	✓
✓	printf("%d", myFunc(2))	0	0	✓
✓	printf("%d", myFunc(10))	1	1	✓
✓	printf("%d", myFunc(25))	0	0	✓
✓	printf("%d", myFunc(200))	1	1	✓

Passed all tests! ✓

Question **2**

Correct

Marked out of 1.00

Flag question

Find the number of ways that a given integer, **X**, can be expressed as the sum of the **Nth** powers of unique, natural numbers.

For example, if **X = 13** and **N = 2**, we have to find all combinations of unique squares adding up to **13**. The only solution is $2^2 + 3^2$.

Function Description

Complete the powerSum function in the editor below. It should return an integer that represents the number of possible combinations.

powerSum has the following parameter(s):



Question **2**

Correct

Marked out of 1.00

Flag question

Find the number of ways that a given integer, X , can be expressed as the sum of the N^{th} powers of unique, natural numbers.

For example, if $X = 13$ and $N = 2$, we have to find all combinations of unique squares adding up to 13 . The only solution is $2^2 + 3^2$.

Function Description

Complete the powerSum function in the editor below. It should return an integer that represents the number of possible combinations.

powerSum has the following parameter(s):

- X: the integer to sum to
 - N: the integer power to raise numbers to
- Input Format

The first line contains an integer X .
The second line contains an integer N .

Constraints

$1 \leq X \leq 1000$
 $2 \leq N \leq 10$

Output Format

Output a single integer, the number of possible combinations calculated.

Sample Input 0

10
2

Sample Output 0

1

Explanation 0

If $X = 10$ and $N = 2$, we need to find the number of ways that 10 can be represented as the sum of squares of unique numbers.

$10 = 1^2 + 3^2$

This is the only way in which 10 can be expressed as the sum of unique squares.

Sample Input 1

100
2

Sample Output 1

3

Explanation 1

If $x = 10$ and $N = 2$, we need to find the number of ways that 10 can be represented as the sum of squares of unique numbers.

$$10 = 1^2 + 3^2$$

This is the only way in which 10 can be expressed as the sum of unique squares.

Sample Input 1

100
2

Sample Output 1

3

Explanation 1

$$100 = (10^2) = (6^2 + 8^2) = (1^2 + 3^2 + 4^2 + 5^2 + 7^2)$$

Sample Input 2

100
3

Sample Output 2

1

Explanation 2

100 can be expressed as the sum of the cubes of $1, 2, 3, 4$.
($1 + 8 + 27 + 64 = 100$). There is no other way to express 100 as the sum of cubes.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2   * Complete the 'powerSum' function below
3   *
4   * The function is expected to return an
5   * The function accepts following paramet
6   * 1. INTEGER x
7   * 2. INTEGER n
8   */
9  #include<math.h>
10 int powerSum(int x, int m, int n)
11 {
12     int p =pow(m,n);
13     if(p==x){
14         return 1;
15     }
16     if(p>x){
17         return 0;
18     }
19     return powerSum(x-p,m+1,n) + powerSum
20
21 }
```

	Test	Expected
✓	printf("%d", powerSum(10, 1, 2))	1

Passed all tests! ✓

Finish review

The index of the pivot is 2.

Sample Case 1

Sample Input 1

STDIN Function Parameters

3 → arr[] size n = 3
1 → arr = [1, 2, 1]
2
1

Sample Output 1

1

Explanation 1

- The first and last elements are equal to 1.
- Using zero based indexing, arr[1]=2 is the pivot between the two subarrays.
- The index of the pivot is 1.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*  
2  * Complete the 'balancedSum' function below.  
3  *  
4  * The function is expected to return an integer.  
5  * The function accepts INTEGER_ARRAY arr as parameter.  
6  */  
7  
8  int balancedSum(int arr_count, int* arr)  
9  {  
10     int left = 0, right = 0;  
11     for(int i=0; i<arr_count; i++)  
12     {  
13         right += arr[i];  
14     }  
15     for(int i=0; i<arr_count; i++){  
16         if(left == (right - arr[i]))  
17             return i;  
18         left += arr[i];  
19         right -= arr[i];  
20     }  
21     return 1;  
22 }  
23  
24  
25
```

	Test	Expected
✓	int arr[] = {1,2,3,3}; printf("%d", balancedSum(4, arr))	2
Passed all tests! ✓		

Question 2

Correct

Flag question

Calculate the sum of an array of integers.

Example

numbers = [3, 13, 4, 11, 9]

The sum is 3 + 13 + 4 + 11 + 9 = 40.

Function Description

Complete the function arraySum in the editor below.

arraySum has the following parameter(s):

int numbers[n]: an array of integers

Correct

Flag question

Example

numbers = [3, 13, 4, 11, 9]

The sum is 3 + 13 + 4 + 11 + 9 = 40.

Function Description

Complete the function arraySum in the editor below.

arraySum has the following parameter(s):
int numbers[n]: an array of integers

Returns

int: integer sum of the numbers array

Constraints

$1 \leq n \leq 10^4$
 $1 \leq \text{numbers}[i] \leq 10^4$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n, the size of the array numbers.
Each of the next n lines contains an integer numbers[i] where $0 \leq i < n$.

Sample Case 0
Sample Input 0

STDIN	Function
5	→ numbers[] size n = 5
1	→ numbers = [1, 2, 3, 4, 5]
2	
3	
4	
5	

Sample Output 0

15

Explanation 0

1 + 2 + 3 + 4 + 5 = 15.
Sample Case 1
Sample Input 1

STDIN	Function
2	→ numbers[] size n = 2
12	→ numbers = [12, 12]
12	

Sample Output 1

24

Explanation 1



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Finish review

Status	Finished
Started	Monday, 13 January 2025, 6:26 PM
Completed	Monday, 13 January 2025, 6:43 PM
Duration	16 mins 30 secs

Question **1**

Correct

Flag question

Given an array of numbers, find the index of the smallest array element (the pivot), for which the sums of all elements to the left and to the right are equal. The array may not be reordered.

Example

arr=[1,2,3,4,6]

- the sum of the first three elements, 1+2+3=6. The value of the last element is 6.
- Using zero based indexing, arr[3]=4 is the pivot between the two subarrays.
- The index of the pivot is 3.

Function Description

Complete the function balancedSum in the editor below.

balancedSum has the following parameter(s):
int arr[n]: an array of integers

Returns:
int: an integer representing the index of the pivot

Constraints

- $3 \leq n \leq 10^5$
- $1 \leq arr[i] \leq 2 \times 10^4$, where $0 \leq i < n$
- It is guaranteed that a solution always exists.

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n, the size of the array arr.
Each of the next n lines contains an integer, arr[i], where $0 \leq i < n$.

Sample Case 0
Sample Input 0

STDIN	Function Parameters
-----	-----

4	→ arr[] size n = 4
1	→ arr = [1, 2, 3, 3]
2	
3	
3	

Sample Output 0

2

Explanation 0

15

Explanation 0

1 + 2 + 3 + 4 + 5 = 15.

Sample Case 1

Sample Input 1

STDIN Function

2 → numbers[] size n = 2

12 → numbers = [12, 12]

12

Sample Output 1

24

Explanation 1

12 + 12 = 24.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2  * Complete the 'arraySum' function below
3  *
4  * The function is expected to return an
5  * The function accepts INTEGER_ARRAY num
6  */
7
8  int arraySum(int numbers_count, int *numb
9  {
10     int sum=0;
11     for(int i=0;i<numbers_count;i++){
12         sum += numbers[i];
13     }
14     return sum;
15 }
16
17
```

	Test	Expected	Got
✓	int arr[] = {1,2,3,4,5}; printf("%d", arraySum(5, arr))	15	15

Passed all tests! ✓

Question 3

Correct

Flag question

Given an array of n integers, rearrange them so that the sum of the absolute differences of all adjacent elements is minimized. Then, compute the sum of those absolute differences. Example n = 5 arr = [1, 3, 3, 2, 4] If the list is rearranged as arr' = [1, 2, 3, 3, 4], the absolute differences are |1 - 2| = 1, |2 - 3| = 1, |3 - 3| = 0, |3 - 4| = 1. The sum of those differences is 1 + 1 + 0 + 1 = 3. Function Description Complete the function minDiff in the editor below. minDiff has the following parameter: arr: an integer array Returns: int: the sum of the absolute differences of adjacent elements Constraints 2 ≤ n ≤ 105 0 ≤ arr[i] ≤ 109, where 0 ≤ i < n Input Format For Custom Testing The first line of input contains an integer, n, the size of arr. Each of the following n lines contains an integer that describes arr[i] (where 0 ≤ i < n) . Sample Case 0 Sample Input For Custom Testing STDIN Function ----- 5 → arr[] size n = 5 5 → arr[] = [5, 1, 3, 7, 3] 1 3 7 3 Sample Output 6 Explanation n = 5 arr = [5, 1, 3, 7, 3] If arr is rearranged as arr' = [1, 3, 3, 5, 7], the differences are minimized. The final answer is |1 - 3| + |3 - 3| + |3 - 5| + |5 - 7| = 6. Sample Case 1 Sample Input For Custom Testing STDIN Function ----- 2 → arr[] size n = 2 3 → arr[] = [3, 2] 2 Sample Output 1 Explanation n = 2 arr = [3, 2] There is no


```
11     for (int i=0; i<numbers_count; i++){
12         sum += numbers[i];
13     }
14     return sum;
15 }
16 }
17 }
```

	Test	Expected	Got
✓	int arr[] = {1,2,3,4,5}; printf("%d", arraySum(5, arr))	15	15

Passed all tests! ✓

Question **3**

Correct

 Flag question

Given an array of n integers, rearrange them so that the sum of the absolute differences of all adjacent elements is minimized. Then, compute the sum of those absolute differences. Example n = 5 arr = [1, 3, 3, 2, 4] If the list is rearranged as arr' = [1, 2, 3, 3, 4], the absolute differences are |1 - 2| = 1, |2 - 3| = 1, |3 - 3| = 0, |3 - 4| = 1. The sum of those differences is 1 + 1 + 0 + 1 = 3. Function Description Complete the function minDiff in the editor below. minDiff has the following parameter: arr: an integer array Returns: int: the sum of the absolute differences of adjacent elements Constraints $2 \leq n \leq 105$ $0 \leq arr[i] \leq 109$, where $0 \leq i < n$ Input Format For Custom Testing The first line of input contains an integer, n, the size of arr. Each of the following n lines contains an integer that describes arr[i] (where $0 \leq i < n$) . Sample Case 0 Sample Input For Custom Testing STDIN Function ---- 5 → arr[] size n = 5 5 → arr[] = [5, 1, 3, 7, 3] 1 3 7 3 Sample Output 6 Explanation n = 5 arr = [5, 1, 3, 7, 3] If arr is rearranged as arr' = [1, 3, 3, 5, 7], the differences are minimized. The final answer is |1 - 3| + |3 - 3| + |3 - 5| + |5 - 7| = 6. Sample Case 1 Sample Input For Custom Testing STDIN Function ---- 2 → arr[] size n = 2 3 → arr[] = [3, 2] 2 Sample Output 1 Explanation n = 2 arr = [3, 2] There is no need to rearrange because there are only two elements. The final answer is |3 - 2| = 1.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2   * Complete the 'minDiff' function below.
3   *
4   * The function is expected to return an
5   * The function accepts INTEGER_ARRAY arr
6   */
7
8  int minDiff(int arr_count, int* arr)
9  {
10     for(int i=0; i<arr_count-1; i++){
11         for(int j=0; j<arr_count-i-1; j++){
12             if(arr[j]>arr[j+1]){
13                 int temp = arr[j];
14                 arr[j]=arr[j+1];
15                 arr[j+1]=temp;
16             }
17         }
18     }
19     int sum=0;
20     for(int i=0; i<arr_count-1; i++){
21         sum += abs(arr[i]-arr[i+1]);
22     }
23     return sum;
24 }
25 }
26 }
```

	Test	Expected	Got
✓	int arr[] = {5, 1, 3, 7, 3}; printf("%d", minDiff(5, arr))	6	6

Passed all tests! ✓



GE23131-Programming Using C-2024

Quiz navigation



Show one page at a time

Finish review

Status	Finished
Started	Monday, 13 January 2025, 6:46 PM
Completed	Monday, 13 January 2025, 7:17 PM
Duration	31 mins 1 sec

Question **1**

Correct

Flag question

You are transporting some boxes through a tunnel, where each box is a parallelepiped, and is characterized by its length, width and height.

The height of the tunnel **41** feet and the width can be assumed to be infinite. A box can be carried through the tunnel only if its height is strictly less than the tunnel's height. Find the volume of each box that can be successfully transported to the other end of the tunnel. Note: Boxes cannot be rotated.

Input Format

The first line contains a single integer ***n***, denoting the number of boxes.

n lines follow with three integers on each separated by single spaces - ***length_i***, ***width_i*** and ***height_i*** which are length, width and height in feet of the ***i***-th box.

Constraints

$1 \leq n \leq 100$

$1 \leq length_i, width_i, height_i \leq 100$

Output Format

For every box from the input which has a height lesser than **41** feet, print its volume in a separate line.

Sample Input 0

4

5 5 5

1 2 40

10 5 41

7 2 42

Sample Output 0

125

80

Explanation 0

The first box is really low, only **5** feet tall, so it can pass through the tunnel and its volume is **5 x 5 x 5 = 125**.

The second box is sufficiently low, its volume is **1 x 2 x 4 = 80**.

The third box is exactly **41** feet tall, so it cannot pass. The same can be said about the fourth box.



125
80

Explanation 0

The first box is really low, only **5** feet tall, so it can pass through the tunnel and its volume is **5 x 5 x 5 = 125**.

The second box is sufficiently low, its volume is **1 x 2 x 4 = 80**.

The third box is exactly **41** feet tall, so it cannot pass. The same can be said about the fourth box.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     for(int i=0;i<n;i++)
7     {
8         int length,width,height;
9         scanf("%d %d %d",&length,&width,&height);
10        if(height<41)
11        {
12            int volume = length*width*height;
13            printf("%d\n",volume);
14        }
15    }
16    return 0;
17 }
```

	Input	Expected	Got	
✓	4	125	125	✓
	5 5 5	80	80	
	1 2 40			
	10 5 41			
	7 2 42			

Passed all tests! ✓

Question **2**
Correct
[Flag question](#)

You are given ***n*** triangles, specifically, their sides ***a_i***, ***b_i*** and ***c_i***. Print them in the same style but sorted by their areas from the smallest one to the largest one. It is guaranteed that all the areas are different.

The best way to calculate a volume of the triangle with sides ***a***, ***b*** and ***c*** is Heron's formula:

$$S = \sqrt{p * (p - a) * (p - b) * (p - c)}$$
 where $p = (a + b + c) / 2$.

Input Format

First line of each test file contains a single integer ***n***. ***n*** lines follow with ***a_i***, ***b_i*** and ***c_i*** on each separated by single spaces.

Constraints

- $1 \leq n \leq 100$**
- $1 \leq a_i, b_i, c_i \leq 70$**
- $a_i + b_i > c_i, a_i + c_i > b_i$ and $b_i + c_i > a_i$**

Output Format

Print exactly ***n*** lines. On each line print **3** integers separated by single spaces, which are ***a_i***, ***b_i*** and ***c_i*** of the corresponding triangle.



$1 \leq a_i, b_i, c_i \leq 70$
 $a_i + b_i > c_i, a_i + c_i > b_i$ and $b_i + c_i > a_i$

Output Format

Print exactly n lines. On each line print **3** integers separated by single spaces, which are a_i, b_i and c_i of the corresponding triangle.

Sample Input 0

3
7 24 25
5 12 13
3 4 5

Sample Output 0

3 4 5
5 12 13
7 24 25

Explanation 0

The square of the first triangle is **84**. The square of the second triangle is **30**. The square of the third triangle is **6**. So the sorted order is the reverse one.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<math.h>
3 #include<stdlib.h>
4 typedef struct {
5     int a,b,c;
6     double area;
7 }triangle;
8 double calculate_area(int a,int b,int c){
9     double p=(a+b+c)/2.0;
10    return sqrt(p*(p-a)*(p-b)*(p-c));
11 }
12 int compare(const void *t1, const void *t2){
13     triangle *tri1=(triangle*)t1;
14     triangle *tri2=(triangle*)t2;
15     if(tri1->area < tri2->area)
16         return -1;
17     if(tri1->area > tri2->area)
18         return 1;
19     return 0;
20 }
21
22 int main() {
23     int n;
24     scanf("%d",&n);
25     triangle triangles[n];
26     for (int i=0;i<n;i++) {
27         int a,b,c;
28         scanf("%d %d %d",&a,&b,&c);
29         triangles[i].a=a;
30         triangles[i].b=b;
31         triangles[i].c=c;
32         triangles[i].area=calculate_area(a,b,c);
33     }
34     qsort(triangles,n,sizeof(triangle),compare);
35     for(int i=0;i<n;i++){
36         printf("%d %d %d\n",triangles[i].a,triangles[i].b,triangles[i].c);
37     }
38     return 0;
39 }
```

	Input	Expected	Got	
✓	3 7 24 25 5 12 13 3 4 5	3 4 5 5 12 13 7 24 25	3 4 5 5 12 13 7 24 25	✓

Passed all tests! ✓



GE23131-Programming Using C-2024

Quiz navigation



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Finish review

Status	Finished
Started	Monday, 13 January 2025, 7:18 PM
Completed	Monday, 13 January 2025, 7:29 PM
Duration	10 mins 37 secs

Question **1**

Correct

Marked out of 1.00

Flag question

Given an array of integers, reverse the given array in place using an index and loop rather than a built-in function.

Example

arr = [1, 3, 2, 4, 5]

Return the array [5, 4, 2, 3, 1] which is the reverse of the input array.

Function Description

Complete the function *reverseArray* in the editor below.

reverseArray has the following parameter(s):

int arr[n]: an array of integers

Return

int[n]: the array in reverse order

Constraints

$1 \leq n \leq 100$

$0 < arr[i] \leq 100$

Input Format For Custom Testing

The first line contains an integer, *n*, the number of elements in *arr*.

Each line *i* of the *n* subsequent lines (where $0 \leq i < n$) contains an integer, *arr[i]*.

Sample Case 0

Sample Input For Custom Testing

5
1
3
2
4
5

Sample Output

5
4
2
3
1

Explanation

The input array is [1, 3, 2, 4, 5], so the reverse of the input array is [5, 4, 2, 3, 1].

Sample Case 1

Sample Input For Custom Testing

4
17
10
21
45
Sample Output
45
21
10
17
Explanation

The input array is [17, 10, 21, 45], so the reverse of the input array is [45, 21, 10, 17].



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array is [45, 21, 10, 17].

Answer: (penalty regime: 0 %)

Reset answer

```

1  /*
2  * Complete the 'reverseArray' function below.
3  *
4  * The function is expected to return an integer array.
5  * The function accepts INTEGER_ARRAY arr and integer arr_count as parameters.
6  */
7
8  /*
9  * To return the integer array from the function call, use 'return' to return the array.
10 * - Store the size of the array to be returned in a variable called 'result_count'.
11 * - Allocate the array statically or dynamically.
12 *
13 * For example,
14 * int* return_integer_array_using_static_allocation(int arr_count) {
15 *     *result_count = 5;
16 *     static int a[5] = {1, 2, 3, 4, 5};
17 *     return a;
18 * }
19 *
20 * int* return_integer_array_using_dynamic_allocation(int arr_count) {
21 *     *result_count = 5;
22 *     int *a = malloc(5 * sizeof(int));
23 *     for (int i = 0; i < 5; i++) {
24 *         *(a + i) = i + 1;
25 *     }
26 *     return a;
27 * }
28 */
29
30 int* reverseArray(int arr_count, int *arr, int *result_count) {
31     *result_count = arr_count;
32     for (int i = 0; i < arr_count/2; i++) {
33         int temp = arr[i];
34         arr[i] = arr[arr_count - i - 1];
35         arr[arr_count - i - 1] = temp;
36     }
37     return arr;
38 }

```

Test



```

int arr[] = {1, 3, 2, 4, 5};
int result_count;
int* result = reverseArray(5, arr, &result_count);
for (int i = 0; i < result_count; i++)
    printf("%d\n", *(result + i));

```

Passed all tests! ✓

Question 2

Correct

Marked out of 1.00

Flag question

An automated cutting machine is used to cut rods into segments. The cutting machine can only hold a rod of *minLength* or more, and it can only make one cut at a time. Given the array *lengths[]* representing the desired lengths of each segment, determine if it is possible to make the necessary cuts using this machine. The rod is marked into lengths already, in the order given.

Example

 $n = 3$ $lengths = [4, 3, 2]$ $minLength = 7$

The rod is initially $sum(lengths) = 4 + 3 + 2 = 9$ units long. First cut off the segment of length $4 + 3 = 7$ leaving a rod $9 - 7 = 2$. Then check that the length 7 rod can be cut into segments of lengths 4 and 3. Since 7 is greater than or equal

Sample Input For Custom Testing

STDIN Function

3 → lengths[] size n = 3
5 → lengths[] = [5, 6, 2]
6
2
12 → minLength= 12

Sample Output

Impossible

Explanation

The uncut rod is $5 + 6 + 2 = 13$ units long. After making either cut, the rod will be too short to make the second cut.

Answer: (penalty regime: 0 %)

Reset answer

```
1  /*
2  * Complete the 'cutThemAll' function below.
3  *
4  * The function is expected to return a string.
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER_ARRAY lengths
7  * 2. LONG_INTEGER minLength
8  */
9
10 /*
11 * To return the string from the function, use the following:
12 *
13 * For example,
14 * char* return_string_using_static_allocation() {
15 *     static char s[] = "static allocation of string";
16 *
17 *     return s;
18 * }
19 *
20 * char* return_string_using_dynamic_allocation() {
21 *     char* s = malloc(100 * sizeof(char));
22 *
23 *     s = "dynamic allocation of string";
24 *
25 *     return s;
26 * }
27 *
28 */
29 char* cutThemAll(int lengths_count, long t=0,i=1;
30     for(int i=0;i<=lengths_count-1;i++)
31     {
32         t+=lengths[i];
33     }
34     do
35     {
36         if(t-lengths[lengths_count-i-1]<minLength)
37         {
38             return "Impossible";
39         }
40         i++;
41     }while(i<lengths_count-1);
42     return "Possible";
43 }
44
45
46
47
```

	Test	Expected Output
✓	long lengths[] = {3, 5, 4, 3}; printf("%s", cutThemAll(4, lengths, 9))	Possible
✓	long lengths[] = {5, 6, 2}; printf("%s", cutThemAll(3, lengths, 12))	Impossible

Passed all tests! ✓

Finish review